Project Overview

Threat Modeling Tool for 6G Non-Terrestrial Networks

Course: Advanced Project in Cyber Security (D7086E)

1. Our Project

- **Problem Statement:** Satellite-based Non-Terrestrial Networks depend on exposed radio links and ground infrastructure. Possible threats include radio interference and unauthorised access to ground administration. A clearly defined network scenario is needed to identify relevant threats and explain their priority.

- **Project Goal**: To design and build a small web application for threat modeling of one LEO NTN access scenario. The scenario includes a rural user device connected through a transparent satellite relay to an NTN gateway and ground gNB and one operator's 5G Core. The application will display a structured threat database using STRIDE categories and ranking based on likelihood and impact. The core application will operate without an LLM or specialised hardware.

- **Supervisor:** Sandeewani Wanigasundara.

2. Our Team

- **Team Members:** John N. Saal and Marius C. Untaru**.**

- **Contributions & Responsibilities:** John will focus on literature review and architecture plus threat identification and the web interface. Marius will focus on the database and API plus ranking and application setup. Both students will review threat records and evaluate the tool and prepare the report and presentations.

- **Learning Objectives:** Develop practical skills in NTN security and STRIDE threat modeling plus risk assessment and database design and web development.

3. How We Will Work Together

- **Methodology**: We will follow an Agile approach and divide the project into small independent deliverables. Work will be organised into two-week sprints. Sprint length can be adjusted to match progress and course deadlines.

- **Meeting Cadence:**
    - Weekly team meetings to review progress and assign sprint tasks.
    - Regular check-ins with supervisor Sandeewani Wanigasundara to review deliverables and discuss technical issues.
    - Additional meetings when urgent questions or blockers arise.

- **Communication & Decision-Making:** We will use Discord for team communication and Zoom or Discord for online meetings. Code and documents will be stored in a private GitHub repository. Major architectural decisions will be agreed together during sprint planning and recorded in writing.

4. Expectations

- **Expectations of Ourselves & Each Other:** We will contribute actively and complete agreed tasks on time. We will share progress and raise problems early. Both students will review each other's work and follow the course requirements for academic integrity.

- **Expectations of Supervisor:** We expect guidance on project scope and the technical approach. We also expect feedback on deliverables and the evaluation method to help us meet the course goals.

5. Managing Challenges

- **Identifying Risks**: We will identify risks early and review them during sprint planning and supervisor check-ins. Main risks include scope growth and limited threat evidence plus implementation delays.

- **Conflict Handling:** We will resolve disagreements through open discussion supported by research and testing. Blockers will be raised early. Unresolved disagreements and proposed scope changes will be discussed with the supervisor.

6. Project Boundaries

Multi-Level Scope Structure
Tier 1 defines the core deliverable. Tiers 2 and 3 are optional extensions that depend on available time.
Tier 1 defines the core deliverable. Tiers 2 and 3 are optional extensions that depend on available time.

- **Tier 1: MVP (Core Foundation)**
  - One bounded LEO NTN scenario covering a rural UE and transparent satellite relay plus gateway and ground gNB connected to one operator's 5G Core interfaces.
  - Limited ground administration included in the model.
  - A documented architecture showing components and interfaces plus assets and data flows and trust boundaries.
  - Threat identification using STRIDE and ranking based on likelihood and impact.
  - A small reviewed threat database with score explanations and supporting sources or clearly labelled assumptions.
  - A simple web application and API with component and STRIDE filters plus CSV export.
  - Evaluation of coverage and scoring consistency plus usability and limitations.

- **Tier 2: Extended Capabilities**
  - Additional reviewed threat records within the same scenario.
  - Docker packaging to simplify installation and setup.
- **Tier 3: Optional LLM Interface**
  - An LLM interface that retrieves existing threat records through the API.
  - The extension must not change threat records or ranking scores. All core functions must remain available without it.

Out of Scope

- MEO and HAPS and UAVs plus inter-satellite links and regenerative payloads.
- Full 5G Core internals and satellite command systems.
- Real satellite hardware and live attacks plus orbital or RF simulation.

- Production deployment and LLM training.

Definition of Done

- The architecture and threat-modeling method are documented and reviewed by both students.
- Threat records have valid mappings and clear score explanations plus evidence or assumption labels.
- The web application and API support the agreed filters and ranking plus CSV export.
- Both students can run the core using documented setup instructions without an LLM or specialised hardware.
- Evaluation results and project limitations are documented.
- The mid-seminar package is ready for 21 October 2026. The final report and presentation are completed by the confirmed course deadlines.

Done: We have a functional web app running in Docker, a populated database of threats, an API for Open WebUI to consume, a written architecture and ranking approach, and a live demo in the 6G NTN case study.

7. Success

The project is considered successful when:

- **Technical Success**: Deliver a working threat-modeling application that meets Tier 1 with reviewed threat records and consistent ranking. Both students can run and demonstrate the core independently.
- **Academic Success:** Meet the course goals through clear problem formulation and a justified method plus documented evaluation and critical reflection.
- **Personal & Team Growth**: Develop practical skills in NTN security and threat modeling plus database and web development. Maintain an effective working relationship throughout the project**.**

Summary

Mid-Project Onboarding Brief: Our team is developing a threat-modeling web application for one transparent LEO NTN access scenario. We use STRIDE to identify threats and likelihood and impact scoring to rank them. We work in Agile sprints with weekly team meetings and supervisor check-ins. Code and documents are stored in a private GitHub repository. The LLM interface is optional.

Team Commitment

By signing this Project Team Charter we confirm that we have jointly discussed and agreed on the project's purpose and scope plus our responsibilities and working practices.

We understand that this charter may be updated by mutual agreement if the project or our circumstances change. Material scope changes will be discussed with the supervisor.

Project Team

Name: Marius C. Untaru	Date: 20/09/2026	Signature: 
Name: John N. Saal	Date: 04/10/2026	Signature: 